Why sparse

Production SPICE uses sparse LU , count nonzeros and inspect L/U structure

The idea

- GOLDEN: nnz=6 on the toy MNA conductance matrix

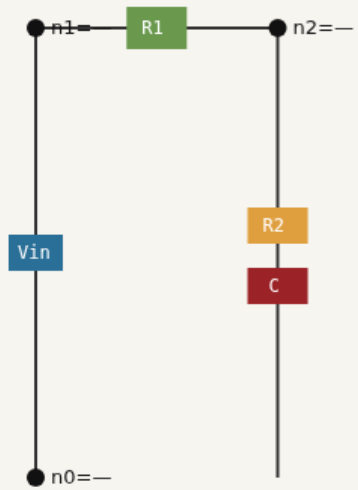
NNZ

SPARSE LU

NNZ

Step 1 / 5 · nnz · module02-03-sparse-lu

nnz=6



Explanation

Count nonzeros in G.

- nnz

6

NNZ

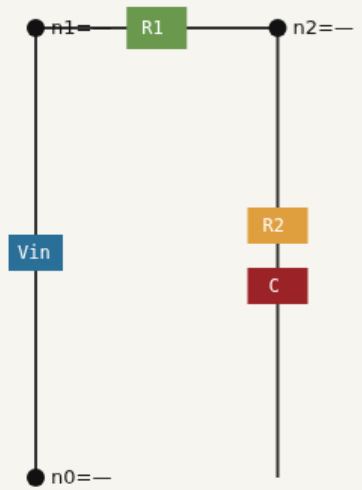
Factor L

SPARSE LU

Factor L

Step 2 / 5 · l · module02-03-sparse-lu

elements=5



Explanation

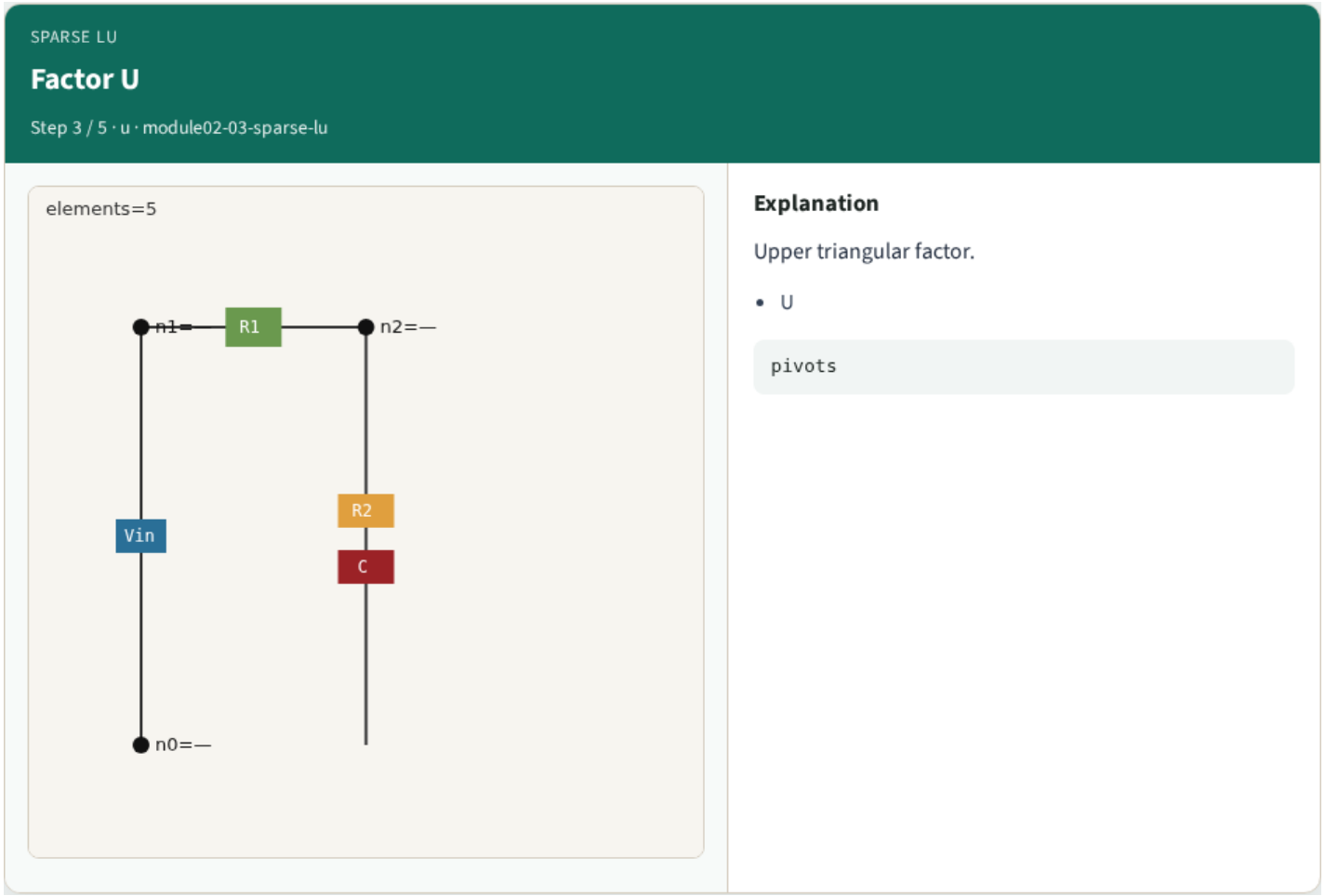
Lower triangular factor.

- L

```
unit diag
```

Factor L

Factor U



Factor U

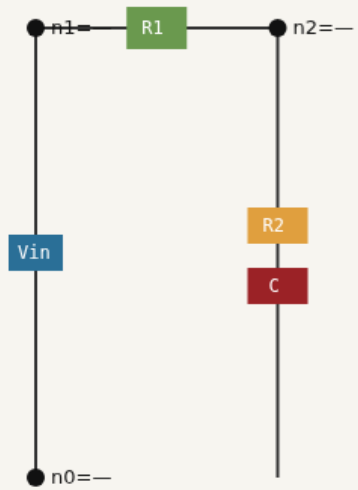
Fwd/back idea

SPARSE LU

Fwd/back idea

Step 4 / 5 · fb · module02-03-sparse-lu

elements=5



Explanation

Solve Ly=b then Ux=y.

- solve

2 sweeps

Fwd/back idea

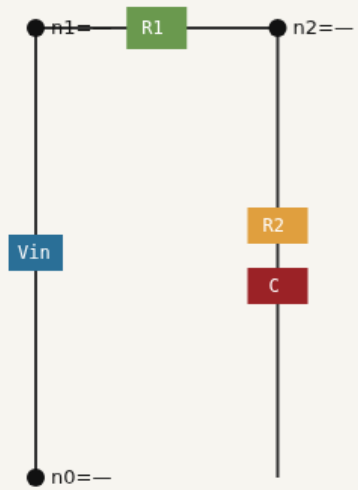
NNZ 6

SPARSE LU

NNZ 6

Step 5 / 5 · n6 · module02-03-sparse-lu

nnz=6



Explanation

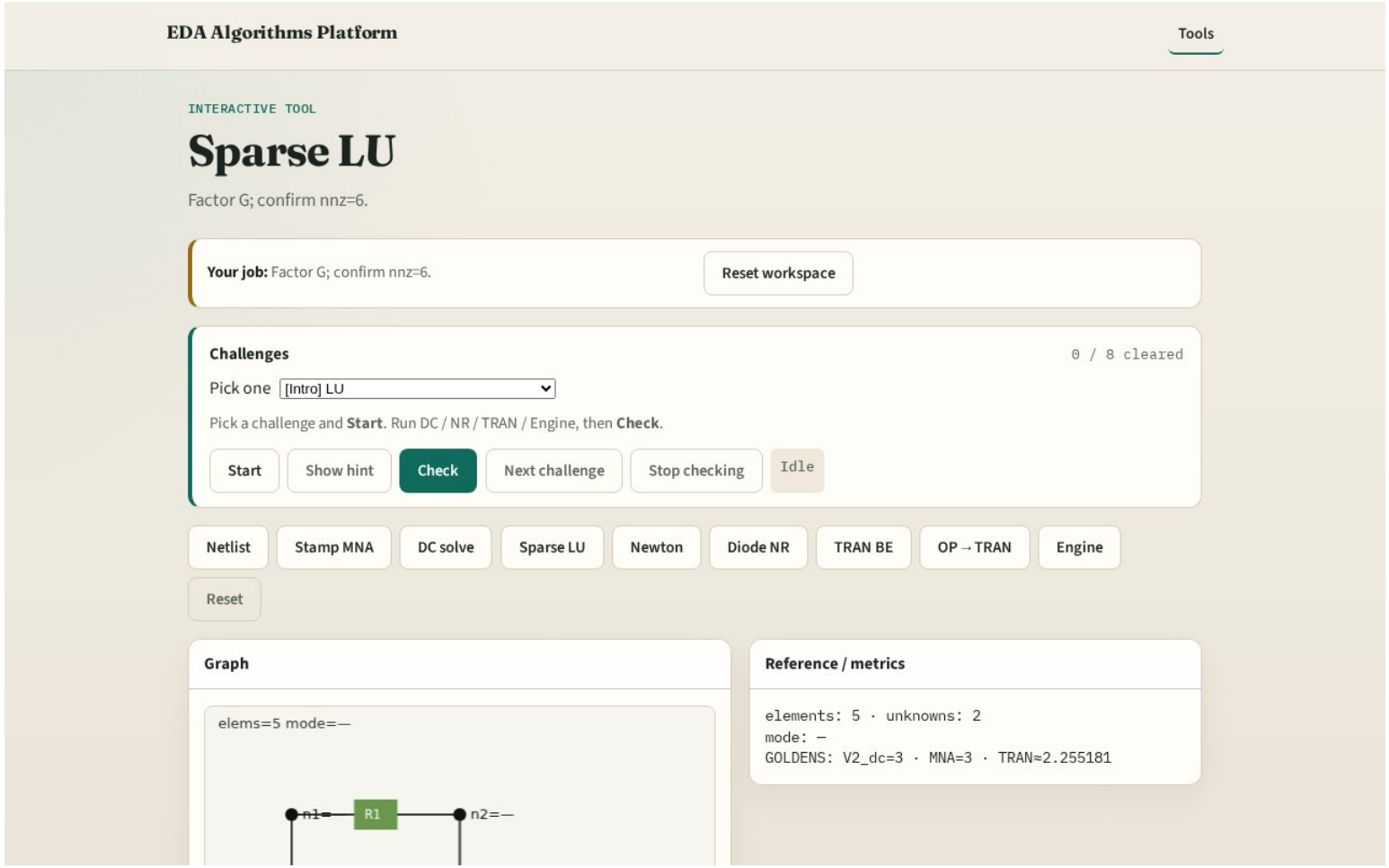
GOLDEN nnz=6 on toy G.

- GOLDEN

```
6
```

NNZ 6

Browser lab track



Browser lab starter

Implement track

- Call sparse_lu_lite(G from dc_linear_solve); assert nnz==6

Pitfalls

- Confusing nnz with matrix dimension
- Expecting fill-in on this tiny dense G

Your turn

- Next: Newton-Raphson on the same stamps